

**Problem**

For some real numbers  $m, n$  and a positive integer  $a$ , the list  $(a+1)n^2, m^2, a(n+1)^2$  consists of three consecutive integers written in increasing order. What is the largest possible value of  $m^2$ ?

Let

$$T_1 = (a+1)n^2, \quad T_2 = m^2, \quad T_3 = a(n+1)^2.$$

Since these are three consecutive integers in increasing order,

$$T_3 - T_1 = 2.$$

Hence

$$a(n+1)^2 - (a+1)n^2 = 2.$$

Expanding,

$$a(n^2 + 2n + 1) - an^2 - n^2 = 2,$$

which simplifies to

$$n^2 - 2an + (2 - a) = 0.$$

Therefore,

$$n = a \pm \sqrt{a^2 + a - 2}.$$

Since the given quantities are integers, we require

$$a^2 + a - 2$$

to be a perfect square. For  $a > 2$ ,

$$(a-1)^2 < a^2 + a - 2 < (a+1)^2,$$

so the only possible perfect square in this interval is  $a^2$ . Thus

$$a^2 + a - 2 = a^2,$$

which gives

$$a = 2.$$

We must also check  $a = 1$ .

**Case 1:**  $a = 1$

Then

$$n = 1,$$

and the three consecutive integers are

$$2, 3, 4.$$

Hence

$$m^2 = 3.$$

**Case 2:**  $a = 2$

Then

$$n = 2 + \sqrt{4} = 4.$$

Thus

$$(a+1)n^2 = 3 \cdot 16 = 48,$$

and the three consecutive integers are

$$48, 49, 50.$$

Hence

$$m^2 = 49.$$

Comparing the two cases, the largest possible value is 49.

**Answer**

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| 49 |
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